EXERCISE 11

A Zero Eigenvalue: Lines of Equilibria

Original statement · probls_ch4-i_2627_v2.pdf, p. 4

Zero Eigenvalue • Line of Equilibria • Parallel Orbits

STATEMENT

Suppose that A has eigenvalues $\lambda_1 = 0, \lambda_2 \neq 0$, with eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$, respectively.

- (i) Write down the general solution of $\dot{\mathbf{x}} = A\mathbf{x}$.
- (ii) After changing to a coordinate system $\tilde{\mathbf{x}}$ referred to the eigenvectors, we obtain $\dfrac{d\tilde{\mathbf{x}}}{dt} = \begin{pmatrix} 0 & 0 \\ 0 & \lambda_2 \end{pmatrix} \tilde{\mathbf{x}}$. By solving these equations draw the phase portrait in the $\tilde{\mathbf{x}}$ system, and sketch the phase portrait for the original coordinates $\mathbf{x}$.
- (iii) Draw the phase portrait for the equation $\dfrac{d\mathbf{x}}{dt} = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix} \mathbf{x}$.

PHASE 1: SETUP AND HYPOTHESES

A is 2×2 real with eigenvalues 0 and $\lambda_2 \neq 0$ (both real: $\lambda_1 = 0$ is real and $\lambda_2 = \operatorname{tr} A$ is real because A is real; being distinct, they have independent eigenvectors; $\det A = 0$). $A\mathbf{v}_1 = \mathbf{0}$, $A\mathbf{v}_2 = \lambda_2 \mathbf{v}_2$. Because $\det A = 0$ the origin is **not** an isolated equilibrium: every point of $\ker A = \operatorname{span}\{\mathbf{v}_1\}$ is an equilibrium ($A\mathbf{x} = \mathbf{0}$).

PHASE 2: FORMULATION

Use the general result for distinct eigenvalues, $\mathbf{x} = c_1 e^{\lambda_1 t}\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2$ (Robinson Eq. 28.1), valid whenever the eigenvalues differ, in particular if one of them is 0; and $Q = [\mathbf{v}_1\ \mathbf{v}_2]$, $\mathbf{z} = Q^{-1}\mathbf{x}$.

PHASE 3: DERIVATION

(i) General solution

With $\lambda_1 = 0$, $e^{\lambda_1 t} = e^0 = 1$:

$$\mathbf{x}(t) = c_1 \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2$$

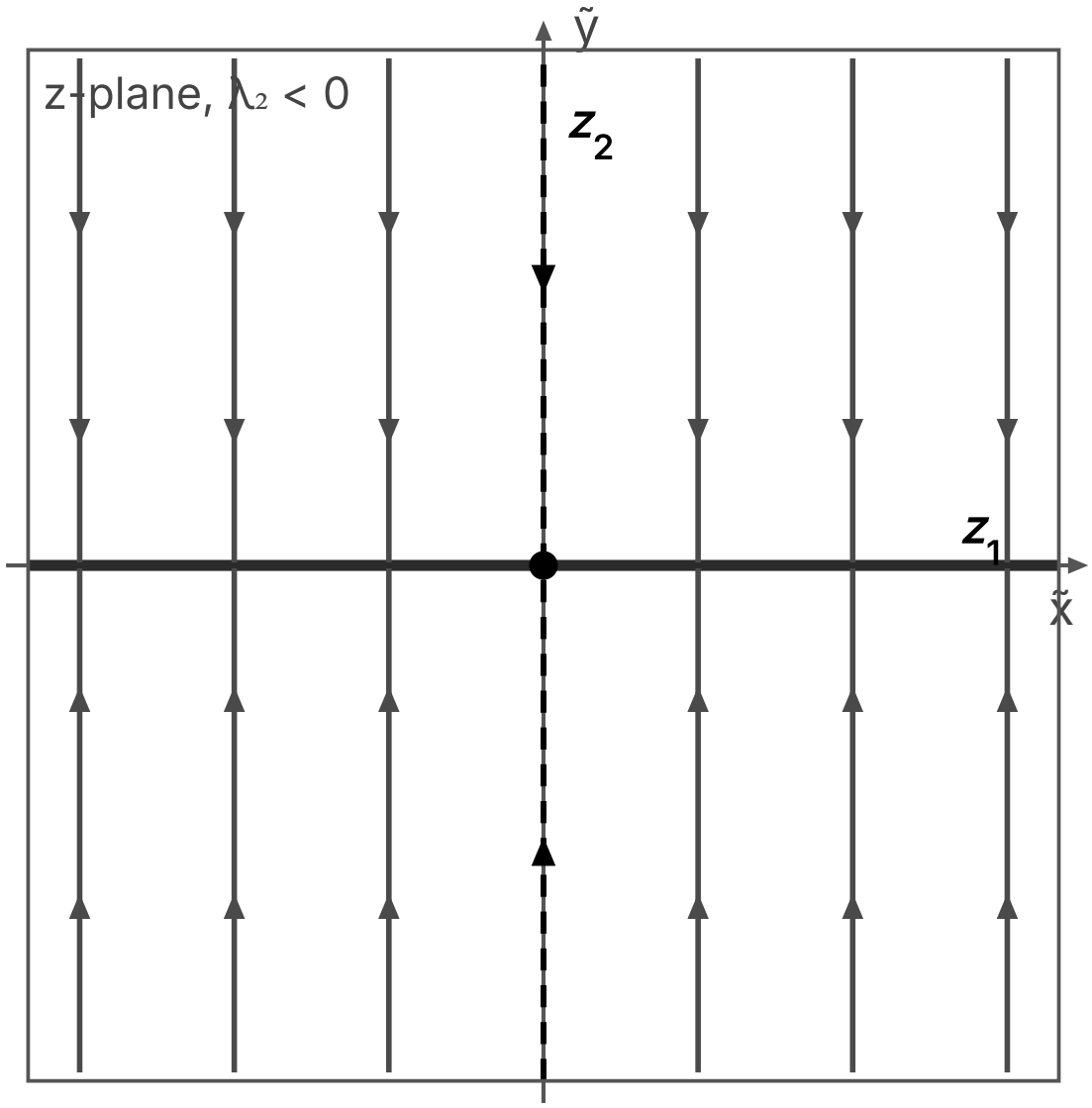
Check: $\dot{\mathbf{x}} = c_2 \lambda_2 e^{\lambda_2 t} \mathbf{v}_2$ and $A\mathbf{x} = c_1 A\mathbf{v}_1 + c_2 e^{\lambda_2 t} A\mathbf{v}_2 = \mathbf{0} + c_2 \lambda_2 e^{\lambda_2 t} \mathbf{v}_2$. They agree. The constant vector $c_1 \mathbf{v}_1$ is an equilibrium and $c_2 e^{\lambda_2 t} \mathbf{v}_2$ the time-dependent part.

(ii) Canonical phase portrait and its image

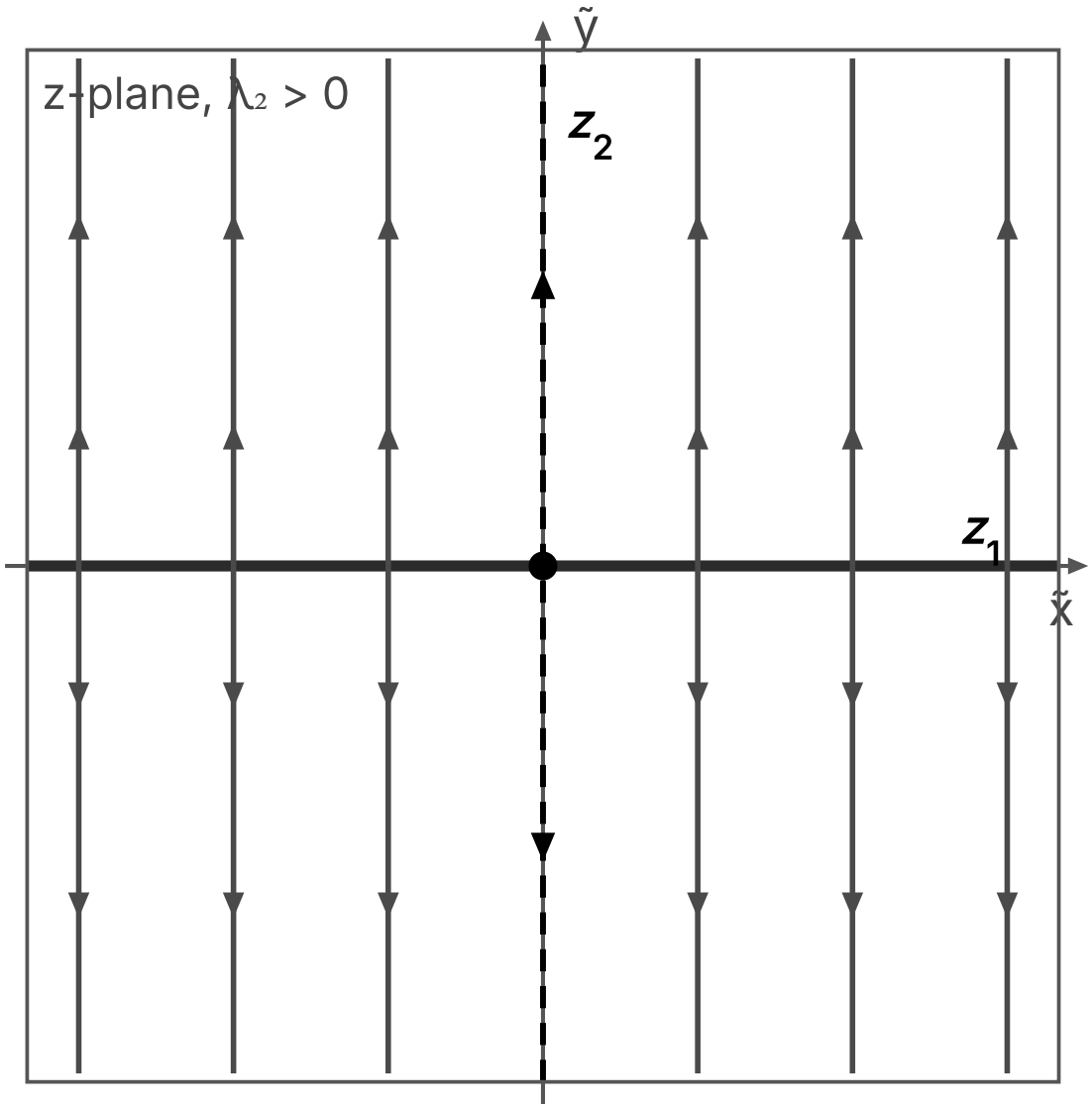
In the coordinates $\tilde{\mathbf{x}} = (\tilde{x}, \tilde{y})^T$ referred to $\mathbf{v}_1, \mathbf{v}_2$ the system is $\dot{\tilde{x}} = 0$, $\dot{\tilde{y}} = \lambda_2 \tilde{y}$. The first gives $\tilde{x} = c_1$ (constant); the second is separable: $\frac{d\tilde{y}}{\tilde{y}} = \lambda_2 \, dt$, $\ln|\tilde{y}| = \lambda_2 t + C$, so $\tilde{y} = c_2 e^{\lambda_2 t}$.

$$\tilde{x}(t) = c_1, \qquad \tilde{y}(t) = c_2 e^{\lambda_2 t}$$

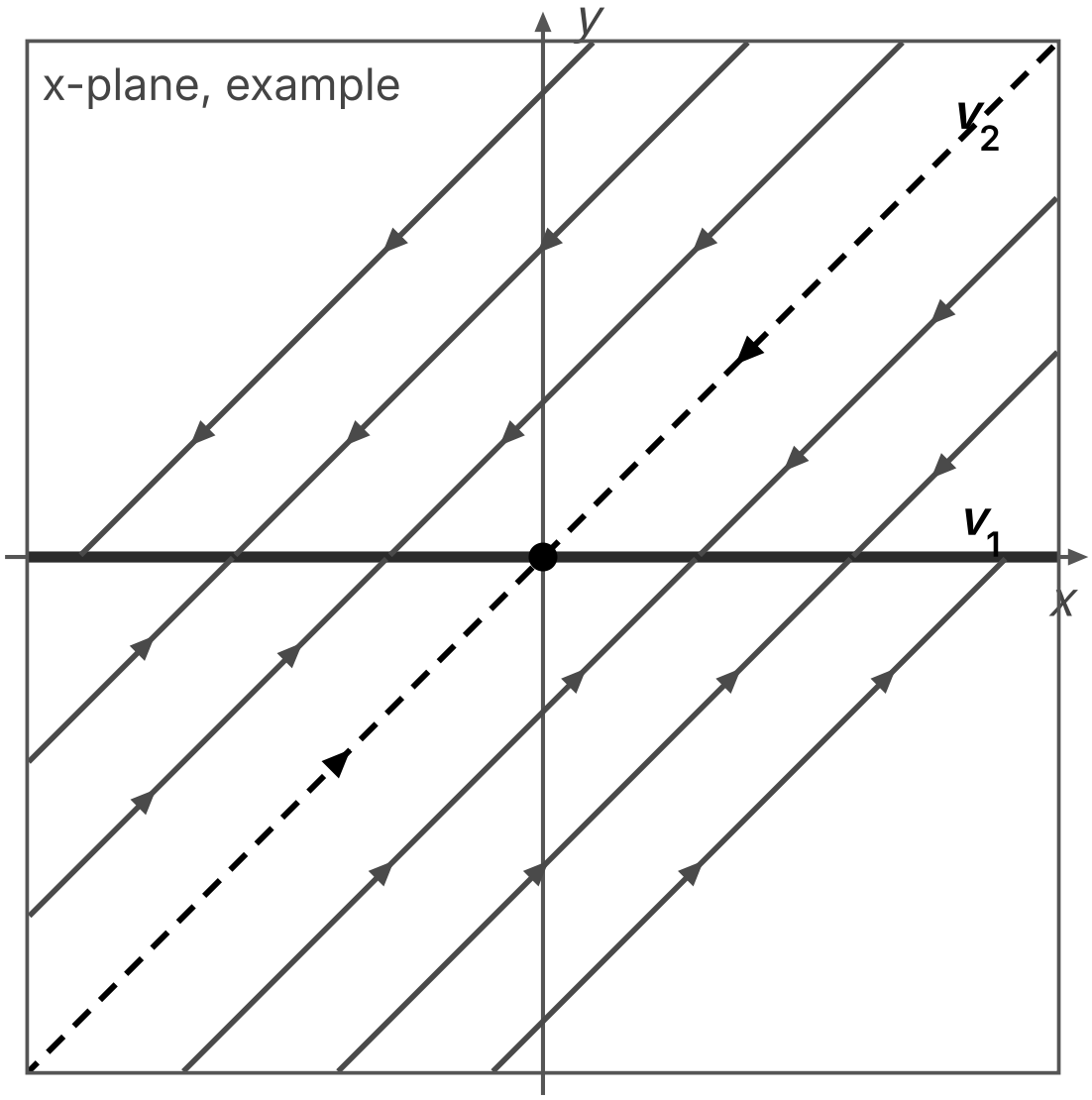
- Orbits: the **vertical lines** $\tilde{x} = c_1$, because $\tilde{x}$ never changes. Each such line is a pair of half-lines (for $\tilde{y} > 0$ and $\tilde{y} < 0$) and the point $(c_1, 0)$ is an equilibrium.
- The $\tilde{x}$-axis ($\tilde{y} = 0$) consists entirely of equilibria (it is the image of $\ker A$).
- Direction: if $\lambda_2 < 0$, $|\tilde{y}| = |c_2| e^{\lambda_2 t}$ decreases: the arrows on every vertical line point **towards** the $\tilde{x}$-axis, which they approach as $t \to +\infty$ without reaching it. If $\lambda_2 > 0$ the arrows point **away** from the axis. The sign of $\dot{\tilde{y}} = \lambda_2 \tilde{y}$ gives the direction at once.
- Original coordinates: $\mathbf{x} = Q\tilde{\mathbf{x}} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$. The $\tilde{x}$-axis goes to the line $\operatorname{span}\{\mathbf{v}_1\}$ of equilibria and the vertical lines go to the lines $c_1\mathbf{v}_1 + s\mathbf{v}_2$, $s \in \mathbb{R}$, **parallel to $\mathbf{v}_2$**, drawn through the points of the line of equilibria. Straight lines remain straight: the picture is the canonical one sheared, with orbits along $\mathbf{v}_2$ towards/away from the line of equilibria.



Canonical $\tilde{\mathbf{x}}$-plane, $\lambda_2 < 0$: the $\tilde{x}$-axis (green) is a line of equilibria and the vertical orbits approach it.



Canonical $\tilde{\mathbf{x}}$-plane, $\lambda_2 > 0$: the vertical orbits leave the line of equilibria.



Original (x, y)-plane for the example $\mathbf{v}_1 = (1, 0)$, $\mathbf{v}_2 = (1, 1)$, $\lambda_2 = -2$: equilibria on the line of $\mathbf{v}_1$, orbits parallel to $\mathbf{v}_2$.

(For the sketch in the original plane, take the example $\mathbf{v}_1 = (1, 0)$, $\mathbf{v}_2 = (1, 1)$, $\lambda_2 = -2$, whose matrix is $A = Q\Lambda Q^{-1} = \begin{pmatrix} 0 & -2 \\ 0 & -2 \end{pmatrix}$; any other eigen-data give the same picture up to a linear change of the axes.)

(iii) The matrix $\begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$

Characteristic equation $\det(A - \lambda I) = 0$ (Topic 4, Eq. 27.8):

$$\det(A - \lambda I) = \begin{vmatrix} -2 - \lambda & 2 \\ 1 & -1 - \lambda \end{vmatrix} = (-2 - \lambda)(-1 - \lambda) - (2)(1)$$

$$= \lambda^2 - (-2 + (-1))\lambda + (-2 \cdot (-1) - 2 \cdot 1) = \lambda^2 - (-3)\lambda + (0) = \lambda^2 + 3\lambda$$

$\operatorname{tr} A = -3$, $\det A = 0$, discriminant $\Delta = (\operatorname{tr} A)^2 - 4 \det A = 9 - 0 = 9$.

$$\lambda = \frac{-3 \pm \sqrt{9}}{2} = \frac{-3 \pm 3}{2} \quad \Longrightarrow \quad \lambda_1 = 0,\ \lambda_2 = -3$$

Check: $\lambda_1 + \lambda_2 = -3 = \operatorname{tr} A$ and $\lambda_1 \lambda_2 = 0 = \det A$.

Eigenvectors

For $\lambda = 0$: $(A - (0)I)\mathbf{v_1} = \mathbf{0}$, that is

$$\begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} -2v_1 + 2v_2 &= 0 \quad \text{(row 1)} \\ 1v_1 - 1v_2 &= 0 \quad \text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A - \lambda I) = 0$), so the rows are proportional (row 2 equals $-\frac{1}{2}$ times row 1) and express a single condition. Row 1 reads $-2v_1 + 2v_2 = 0$, so $v_1 = v_2$: every eigenvector is a multiple of $(1, 1)^T$. The smallest integer representative is

$$\mathbf{v_1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Check: $A\mathbf{v_1} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0\mathbf{v_1}$.

For $\lambda = -3$: $(A - (-3)I)\mathbf{v_2} = \mathbf{0}$, that is

$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} 1v_1 + 2v_2 &= 0 \quad \text{(row 1)} \\ 1v_1 + 2v_2 &= 0 \quad \text{(row 2)} \end{aligned}$$

The coefficient matrix is singular ($\det(A-\lambda I)=0$), so the rows are proportional (row 2 is identical to row 1) and express a single condition. Row 1 reads $1v_1+2v_2=0$, so $v_1=-2\,v_2$: every eigenvector is a multiple of $(-2,1)^T$. The smallest integer representative is

$$\mathbf{v_2}=\begin{pmatrix}2\\-1\end{pmatrix}$$

Check: $A\mathbf{v_2}=\begin{pmatrix}-6\\3\end{pmatrix}=-3\mathbf{v_2}$.

Change of coordinates

The eigenvectors of distinct eigenvalues are linearly independent (Topic 4, Appendix B argument), so they form a basis. Put them as columns, $Q=[\mathbf{v}_1\mid\mathbf{v}_2]$, and define the new coordinates by $\mathbf{z}=Q^{-1}\mathbf{x}$, i.e. $\mathbf{x}=Q\mathbf{z}$:

$$Q=\begin{pmatrix}1&2\\1&-1\end{pmatrix}$$

$\det Q=(1)(-1)-(2)(1)=-3$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix}a&b\\c&d\end{pmatrix}^{-1}=\frac{1}{ad-bc}\begin{pmatrix}d&-b\\-c&a\end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1}=\frac{1}{-3}\begin{pmatrix}-1&-2\\-1&1\end{pmatrix}=\begin{pmatrix}\frac{1}{3}&\frac{2}{3}\\\frac{1}{3}&-\frac{1}{3}\end{pmatrix}$$

Since Q is constant, $\dot{\mathbf{z}}=Q^{-1}\dot{\mathbf{x}}=Q^{-1}A\mathbf{x}=(Q^{-1}AQ)\mathbf{z}$. Compute AQ one column at a time, with $A\mathbf{v}_j=\lambda_j\mathbf{v}_j$:

$$A\mathbf{v}_1=\begin{pmatrix}-2&2\\1&-1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}=0\mathbf{v}_1,\qquad A\mathbf{v}_2=\begin{pmatrix}-2&2\\1&-1\end{pmatrix}\begin{pmatrix}2\\-1\end{pmatrix}=\begin{pmatrix}-6\\3\end{pmatrix}=(-3)\mathbf{v}_2$$

$$AQ=[A\mathbf{v}_1\mid A\mathbf{v}_2]=\begin{pmatrix}0&-6\\0&3\end{pmatrix}$$

Now multiply by Q^{-1} on the left, entry by entry:

$$
\begin{aligned}
(Q^{-1}AQ)_{11} &= \frac{1}{3}\cdot 0 + \frac{2}{3}\cdot 0 = 0 \\
(Q^{-1}AQ)_{12} &= \frac{1}{3}\cdot (-6) + \frac{2}{3}\cdot 3 = 0 \\
(Q^{-1}AQ)_{21} &= \frac{1}{3}\cdot 0 + (-\frac{1}{3})\cdot 0 = 0 \\
(Q^{-1}AQ)_{22} &= \frac{1}{3}\cdot (-6) + (-\frac{1}{3})\cdot 3 = -3
\end{aligned}
$$

$$
Q^{-1}AQ = \begin{pmatrix} 0 & 0 \\ 0 & -3 \end{pmatrix} = \Lambda = \begin{pmatrix} 0 & 0 \\ 0 & -3 \end{pmatrix}
$$

The transformation has decoupled the system (the matrix is diagonal), as claimed.

Solution in the canonical plane

$$
\dot{z}_1 = 0, \qquad \dot{z}_2 = -3z_2
$$

Each equation is separable. If $z_i \neq 0$, $\frac{dz_i}{z_i} = \lambda_i\, dt$ and integrating, $\ln|z_i| = \lambda_i t + C_i$, so $z_i = c_i e^{\lambda_i t}$ with $c_i = \pm e^{C_i}$ (the solution $z_i \equiv 0$ corresponds to $c_i = 0$). Equivalently, $\frac{d}{dt}(e^{-\lambda_i t} z_i) = 0$.

$$
z_1(t) = c_1 e^{0t}, \qquad z_2(t) = c_2 e^{-3t}
$$

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{v}_1 + z_2\mathbf{v}_2$, so the general solution is

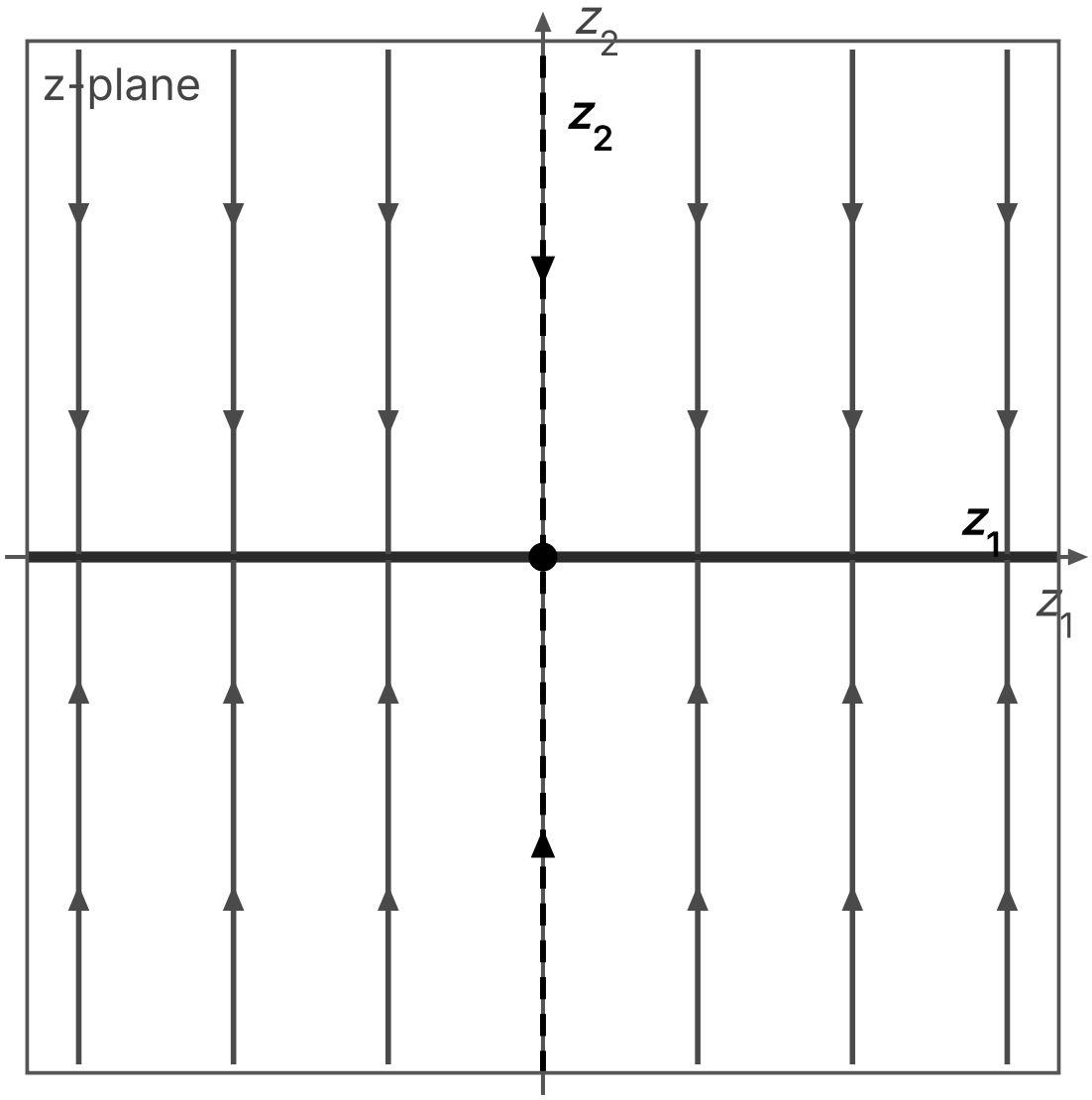
$$
\mathbf{x}(t) = c_1 e^{0t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-3t}\begin{pmatrix} 2 \\ -1 \end{pmatrix}
$$

$$
x(t) = c_1 + 2c_2 e^{-3t}, \qquad y(t) = c_1 - c_2 e^{-3t}
$$

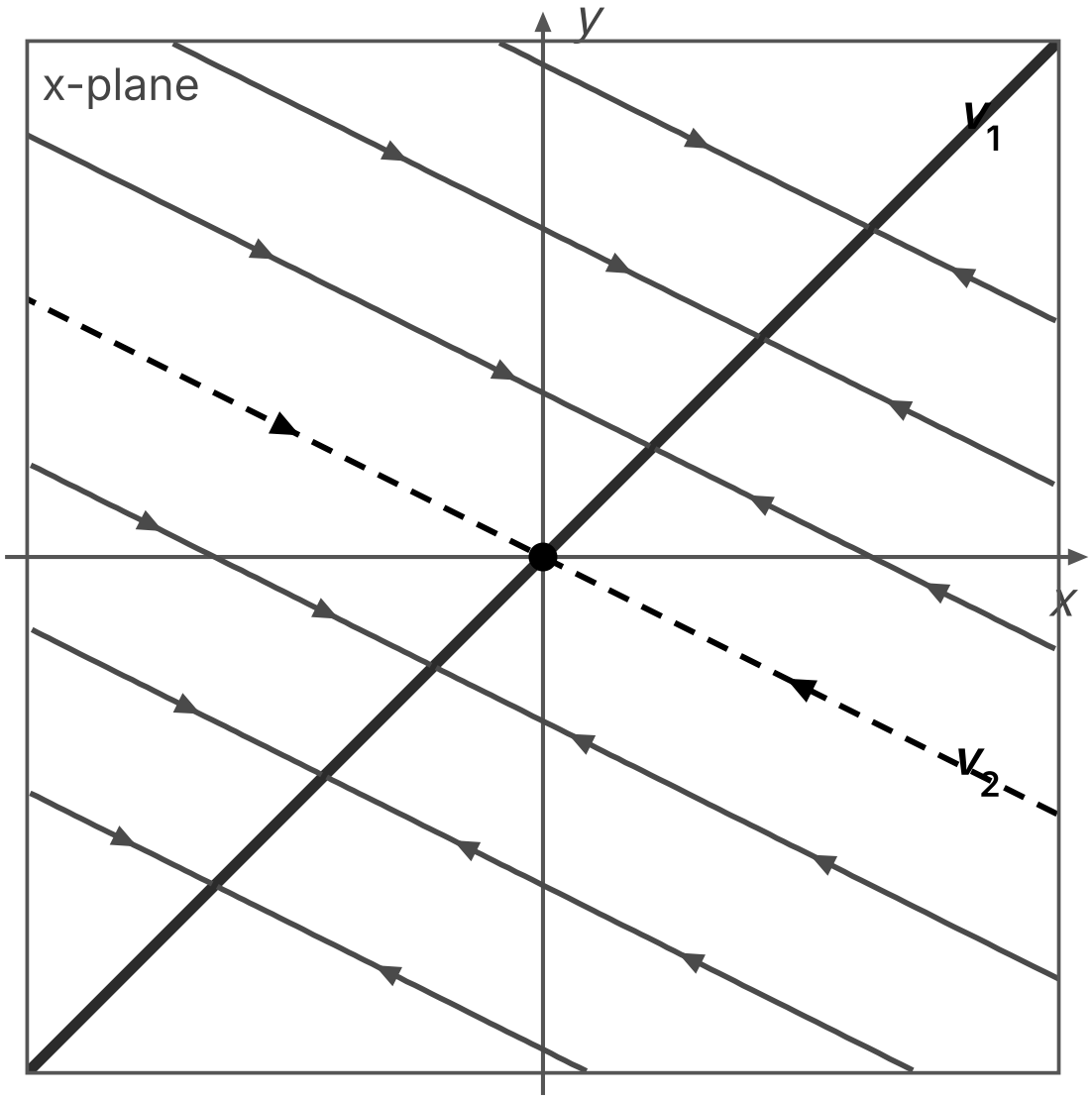
Phase portrait

One eigenvalue is zero and the other is $\lambda_2 = -3 \neq 0$. In the canonical plane $\dot{z}_1 = 0$, so z_1 is constant along every solution (Topic 4, case $\lambda = 0$).

Every point of the line through $\mathbf{v}_1 = (1, 1)$ is an equilibrium, because $A\mathbf{v}_1 = \mathbf{0}$. Every other solution moves on a straight line parallel to $\mathbf{v}_2 = (2, -1)$ with $\mathbf{x} - \mathbf{x}_{eq} \propto e^{-3t}$: it approaches the line of equilibria as $t \to +\infty$ (the equilibria are stable but each orbit ends at a different point). The arrows point towards the line of equilibria.



Canonical $\mathbf{z}$-plane: the line of equilibria (green) and the trajectories parallel to the other axis.



Original (x, y)-plane: equilibria on the line through $(1, 1)$ (green); trajectories parallel to $(2, -1)$, towards that line.

PHASE 4: INTERPRETATION AND CHECKS

- (iii): $\operatorname{tr} A = -3$ and $\det A = (-2)(-1) - 2 \cdot 1 = 0$, so the eigenvalues are 0 and -3: the general solution is $\mathbf{x} = c_1(1, 1)^T + c_2 e^{-3t}(2, -1)^T$, every orbit converges to the equilibrium $c_1(1, 1)^T$, i.e. $\mathbf{x}(t) \to \mathbf{x}_\infty$ on the line $y = x$ along the direction $(2, -1)$.
- Explicit limit for $\mathbf{x}(0) = (x_0, y_0)$: $\mathbf{x}_\infty = c_1(1, 1)^T$ with $c_1 = (x_0 + 2y_0)/3$ (from Q^{-1} computed above, first row $(\frac{1}{3}, \frac{2}{3})$). Example: $\mathbf{x}(0) = (2, -1)$ gives $c_1 = 0$ and the orbit is the line $\mathbf{x} = e^{-3t}(2, -1)$ going to the origin.
- The origin is no longer an isolated equilibrium: the node/saddle/spiral classification of Topic 4 Sec. 9 assumes $\det A \neq 0$ (see the trace-determinant discussion), and here $\det A = 0$, which is the borderline between a stable node ($\lambda_2 < 0$) and a saddle.

- Conserved quantity: the combination $\tilde{x} = c_1$ is a first integral of the motion ($x + 2y$ in (iii), up to the factor $\frac{1}{3}$): $\frac{d}{dt}(x + 2y) = (-2x + 2y) + 2(x - y) = 0$ ✓.